

The Power of Vaccination

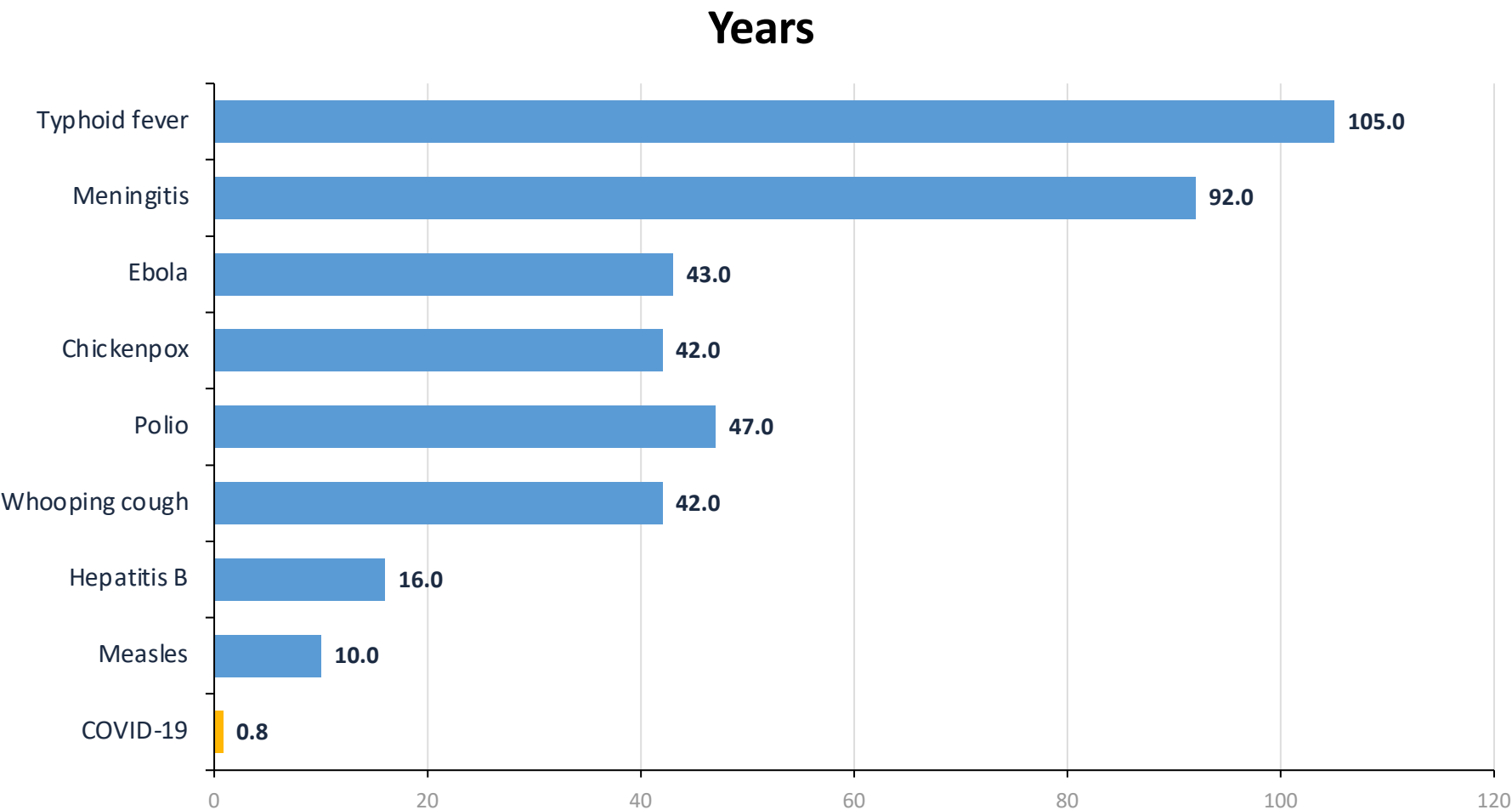
From Smallpox Eradication to COVID-19 in Record Time

A Data-Driven Overview | 1796 – 2020

Source: Our World in Data — <https://ourworldindata.org/vaccination>

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure (years)



COVID-19

< 1 Year

mRNA technology enabled
vaccine development in
record time

**Typhoid: 105 yrs
COVID-19: <1 yr**

130× faster

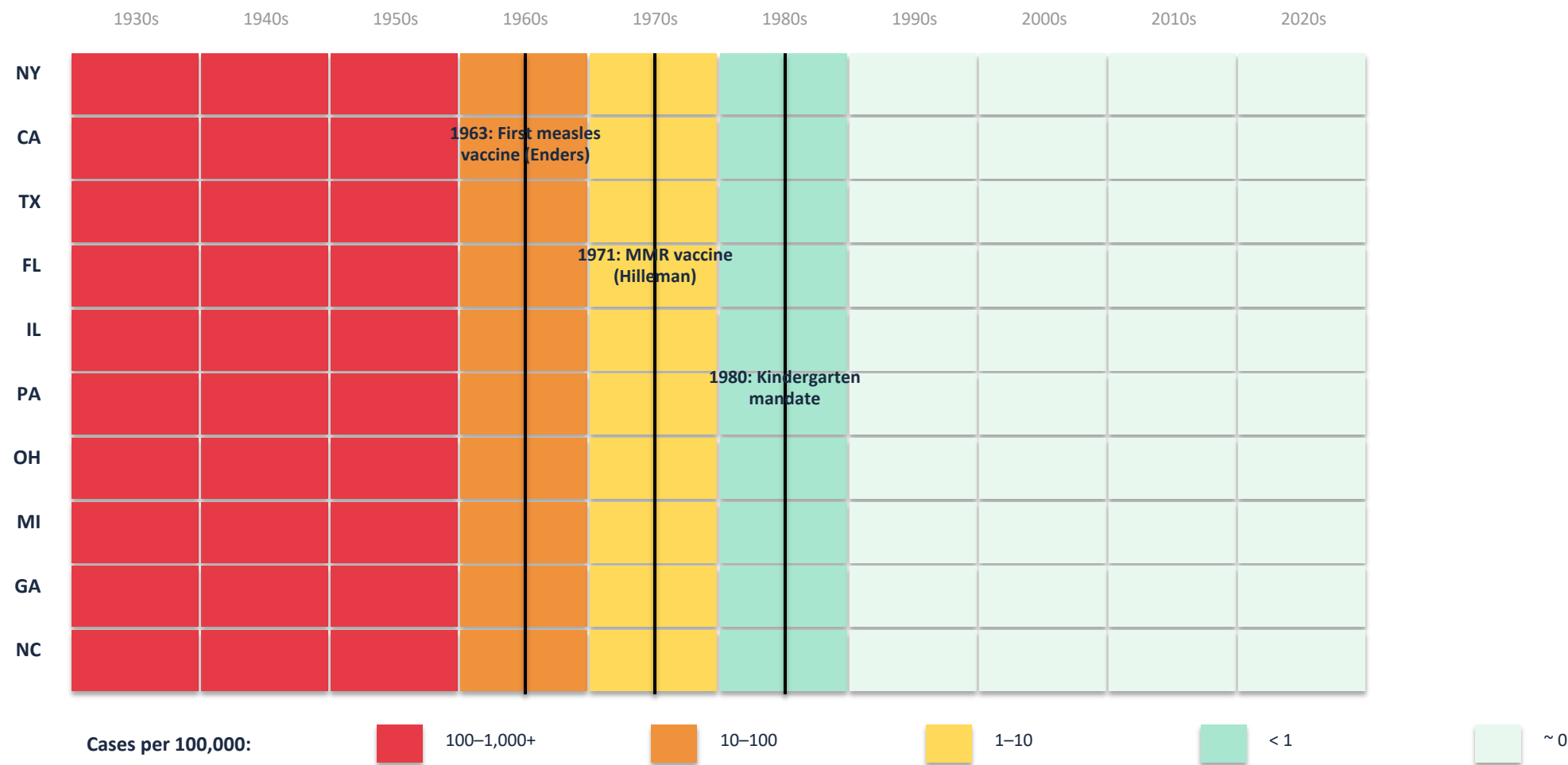
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases/million/yr	Post-Vaccine Cases/million/yr	Case Reduction	Pre-Vaccine Deaths/million/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hep. B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

■ Green rows: 100% case AND death elimination achieved

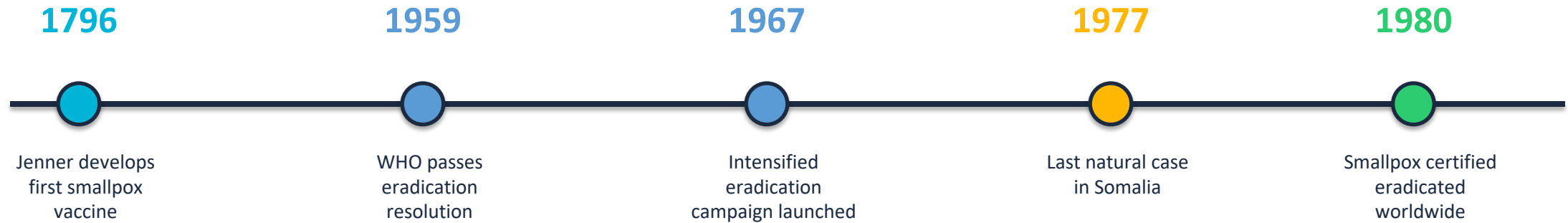
■ Red values: Incomplete reduction — gaps remain

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Schematic heatmap based on state-level measles case data 1929–2022. Before 1963, nearly every state had 100–1,000+ cases per 100,000. After the 1980 mandate, cases dropped to near zero with only sporadic outbreaks.

Smallpox: The Only Human Disease Eradicated Through Vaccination



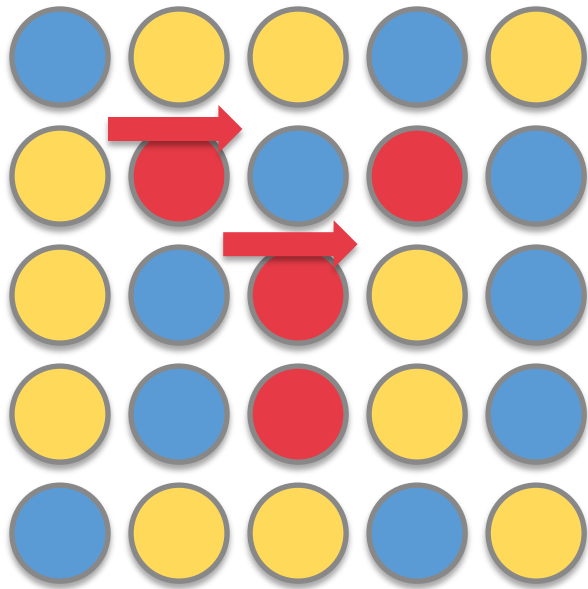
Why was smallpox uniquely suited for eradication?

- The virus infected only humans — no animal reservoir to reignite transmission
- Symptoms were visually distinguishable — cases could be identified and isolated rapidly
- The vaccine was cheap, stable, and highly effective — enabling mass global campaigns

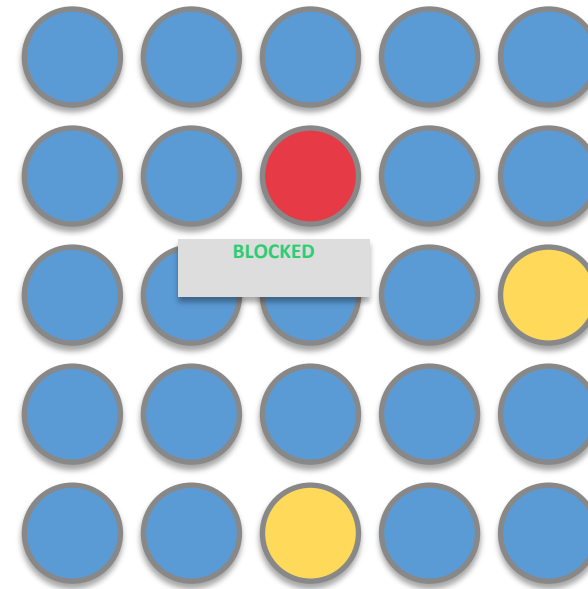
Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively through a population. This shields those who cannot be vaccinated — newborns, immunocompromised individuals, and cancer patients on chemotherapy.

Without Herd Immunity



With Herd Immunity



Vaccinated individual



Vulnerable (cannot be vaccinated)



Infected

Key insight: Vaccines protect not just individuals but entire communities. High vaccination rates create a barrier that stops pathogen spread, safeguarding those most at risk.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal Disease

40.5%

case reduction

31.3%

death reduction

Hepatitis A

89%

case reduction

88.7%

death reduction

Acute Hepatitis B

83.9%

case reduction

83.6%

death reduction

Key Barriers to Full Vaccine Impact

- Access gaps — Not everyone globally has access to vaccines, particularly in low-income regions
- Supply shortages — Vaccine supply disruptions put children at risk of missing scheduled doses
- Trust challenges — Poor scientific understanding and misinformation can reduce vaccine uptake

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

1 **100% elimination** Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US

2 **130× faster** Development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA and genomic tools

3 **Near-zero measles** Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate

4 **Only one eradicated** Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign

5 **Gaps persist** Pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%) show incomplete control

CALL TO ACTION FOR GLOBAL HEALTH SYSTEMS

Improve access · Expand coverage · Strengthen trust · Invest in next-generation vaccines